

Question Number	Answer	Additional guidance	Mark
6(a)(i)	The only correct answer is A <i>B is not correct because Y is reduced NAD and Z is pyruvate</i> <i>C is not correct because W is oxidised NAD and Y is reduced NAD</i> <i>D is not correct because W is oxidised NAD, Y is reduced NAD and Z is pyruvate</i>		(1)

Question Number	Answer	
* 6(a)(ii) 1 and 2 3 4 5 6	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. • blood lactate increases with increasing intensity of exercise • stroke volume increases with increasing heart rate • positive correlation between heart rate and stroke volume for athlete / little change in stroke volume for non-athlete as heart rate increases/ stroke volume always higher in athlete than non-athlete • non athletes started at a higher lactate concentration than athletes/ converse / athletes lactate always lower in non-athletes • any use of data from lactate concentration table • any use of data from stroke volume graph • blood lactate levels increase because there is anaerobic respiration occurring / insufficient oxygen available for aerobic respiration / oxygen stores used up • anaerobic respiration produces lactate / lactate converted to pyruvate / glucose → lactic acid + energy / {anaerobic respiration / lactate} causes blood pH to drop / less anaerobic respiration gives less lactate • stroke volume increases more in athlete due to {(more / stronger) {heart / cardiac} muscle present/ greater ventricular contraction force/ increased contraction of cardiac muscle} /converse • (increase in stroke volume) supplying more (oxygen / glucose) to respiring cells • athletes have larger lung capacity {for oxygen exchange due to training/ more effective gas exchange /to respire aerobically for longer} / converse. • endurance athletes have more {slow twitch muscles /stores of myoglobin / mitochondria} / sprinters will have more fast twitch muscle fibres / non athlete have {greater proportion of / less} fast twitch muscles in their musculature / accept a correct comment about slow / fast muscle fibres • in non-athletes (because stroke volume doesn't increase) less oxygen being delivered to the respiring muscle cells than in athlete / accept converse • non athletes develop larger oxygen debt than athletes (because lactate levels are higher) • drop in blood pH detected by chemoreceptors which start chain of events leading to increase in {ventilation /breathing / heart} rate	

7	- no {error bars / SDs } on stroke volume graph so cannot comment on the significant difference - no details about {age / gender / health / diet} of athletes and non-athletes in investigation (6)		
			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.	aspect = graph or athlete/non-athlete 1 mark = description of one aspect (graph or person) 2 marks = description of both aspects or description of one with some explanation
Level 2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning, with some structure.	3 marks = description of both aspects with some explanation for one 4 marks = description of both aspects with explanation
Level 3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.	All level 2 content plus: consideration of why there is a difference in stroke volume between athlete/non-athlete and why the lactate increases are different between athlete/non-athlete

Question Number	Answer	Additional guidance	Mark
6(b)	An answer that includes four of the following points: • {cytochrome oxidase cannot be oxidised / electrons are not passed along electron transfer chain / reduced {NAD / FAD / coenzyme} are not oxidised} (1) • cells {respire anaerobically / only do glycolysis} (1) • {insufficient / less / no} {ATP / energy} for {contraction of muscles / contraction of heart / metabolic reactions} (1) • pyruvate converted to {lactate / lactic acid} (1) • (accumulation of) {lactate / lactic acid} results in a decreased pH (1) • resulting in denaturing of enzymes / change in {secondary / tertiary} structure (of proteins) (1)	ACCEPT {preventing / less} {oxidative phosphorylation / aerobic respiration} ACCEPT chemiosmosis does not occur ACCEPT {less protons / protons do not} move through ATP synthase ACCEPT examples of muscles e.g. {skeletal / intercostal / diaphragm} muscles ACCEPT named metabolic reaction	(4)

